Day 3 · Results pack

The real tables and charts from the teacher's run of Notebook 2, the Seasonal Workbook (LANE = "halloween costume"). Use them if you fall behind: every number here is what your notebook prints too.

```
Data version: day3-seasonal-v3 (read Oct 29, 2026)
Fingerprint: af2cf27c52e5 (matches the answer sheet)
Trends: 2096 rows × 3 columns, from the built-in copy
Spending: 22 rows × 16 columns
```

What's inside

1. Summary tables: what is in the data, the seasonal calendar, the peaks
2. The seven gallery charts, each with the question it answers and what we can infer
3. The honest-axis pair
4. Season maths for every lane
5. The halloween costume lane (section 6): the four charts and the two small tables for an AI-built dashboard

Sources

Google Trends: Canada, web search, weekly, Oct 24, 2021 to Oct 25, 2026, one search term per lane; relative interest 0 to 100 per lane (not counts, not sales); read Oct 29, 2026. NRF Halloween Data Center: the National Retail Federation's annual Halloween consumer survey by Prosper Insights & Analytics, 2005 to 2026; US consumers, US dollars; each year's figure is the September forecast. Partial years are flagged, never hidden.

1 · Summary tables

Each lane described (262 weeks each)

	count	mean	std	min	25%	50%	75%	max
lane								
back to school	262.0	18.6	21.8	5.0	7.0	10.0	16.8	100.0
candy	262.0	46.6	11.6	34.0	39.0	42.5	50.0	100.0
christmas gifts	262.0	13.6	25.6	0.0	1.0	2.0	8.0	100.0
halloween costume	262.0	8.6	17.2	1.0	1.0	1.0	5.0	100.0
pumpkin spice	262.0	17.0	23.7	0.0	3.0	4.0	17.0	100.0
snow tires	262.0	19.4	15.8	3.0	8.0	13.0	28.8	100.0
sunscreen	262.0	39.7	21.1	8.0	23.0	33.0	54.8	100.0
valentines day	262.0	4.4	14.1	0.0	1.0	1.0	1.0	100.0

The seasonal calendar: average interest by lane and month

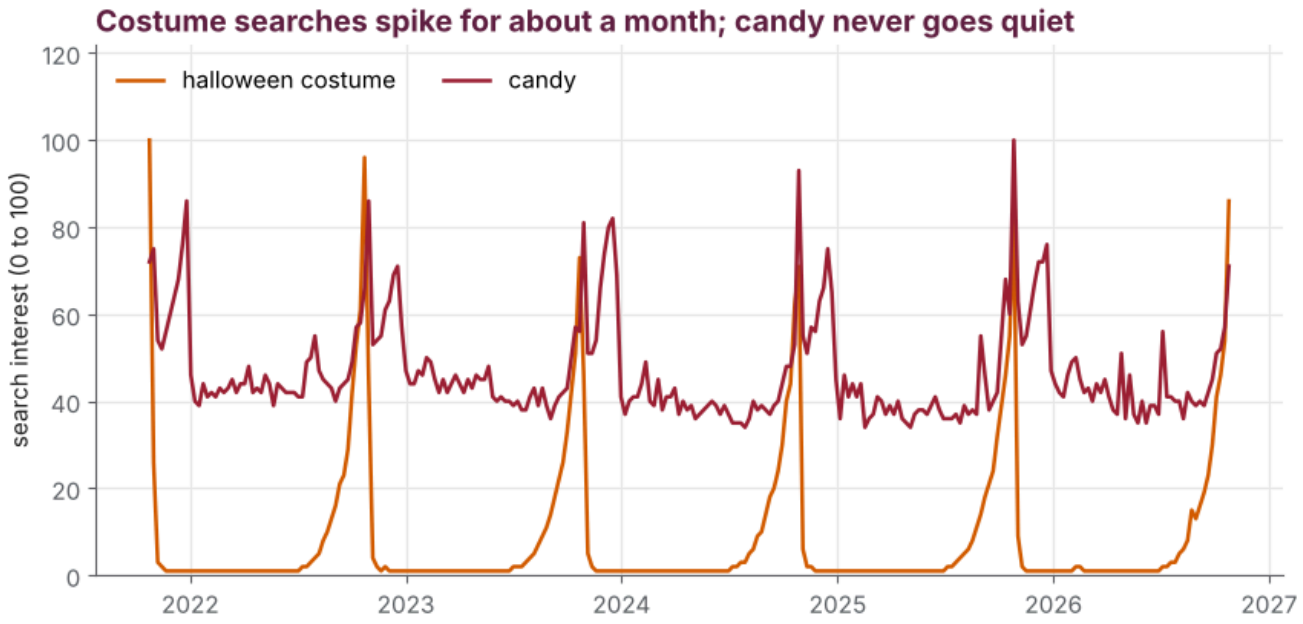
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
lane												
halloween costume	1.0	1.0	1.0	1.0	1.0	1.0	3.0	9.0	22.0	56.0	2.0	1.0
candy	43.0	43.0	42.0	41.0	40.0	39.0	41.0	41.0	41.0	63.0	56.0	67.0
pumpkin spice	4.0	4.0	3.0	3.0	2.0	2.0	5.0	39.0	63.0	55.0	14.0	8.0
christmas gifts	2.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	5.0	17.0	62.0	67.0
valentines day	14.0	35.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	1.0	2.0
sunscreen	25.0	31.0	41.0	54.0	73.0	68.0	61.0	40.0	25.0	19.0	20.0	19.0
snow tires	20.0	14.0	12.0	9.0	6.0	5.0	7.0	12.0	24.0	42.0	50.0	32.0
back to school	16.0	8.0	8.0	8.0	9.0	15.0	25.0	73.0	30.0	11.0	7.0	13.0

Halloween costume: the busiest week of each year

	lane	year	week	interest
0	halloween costume	2021	2021-10-24	100
52	halloween costume	2022	2022-10-23	96
104	halloween costume	2023	2023-10-22	73
157	halloween costume	2024	2024-10-27	71
209	halloween costume	2025	2025-10-26	82
261	halloween costume	2026	2026-10-25	86

Chart 1

Line: when does Halloween start on Google?

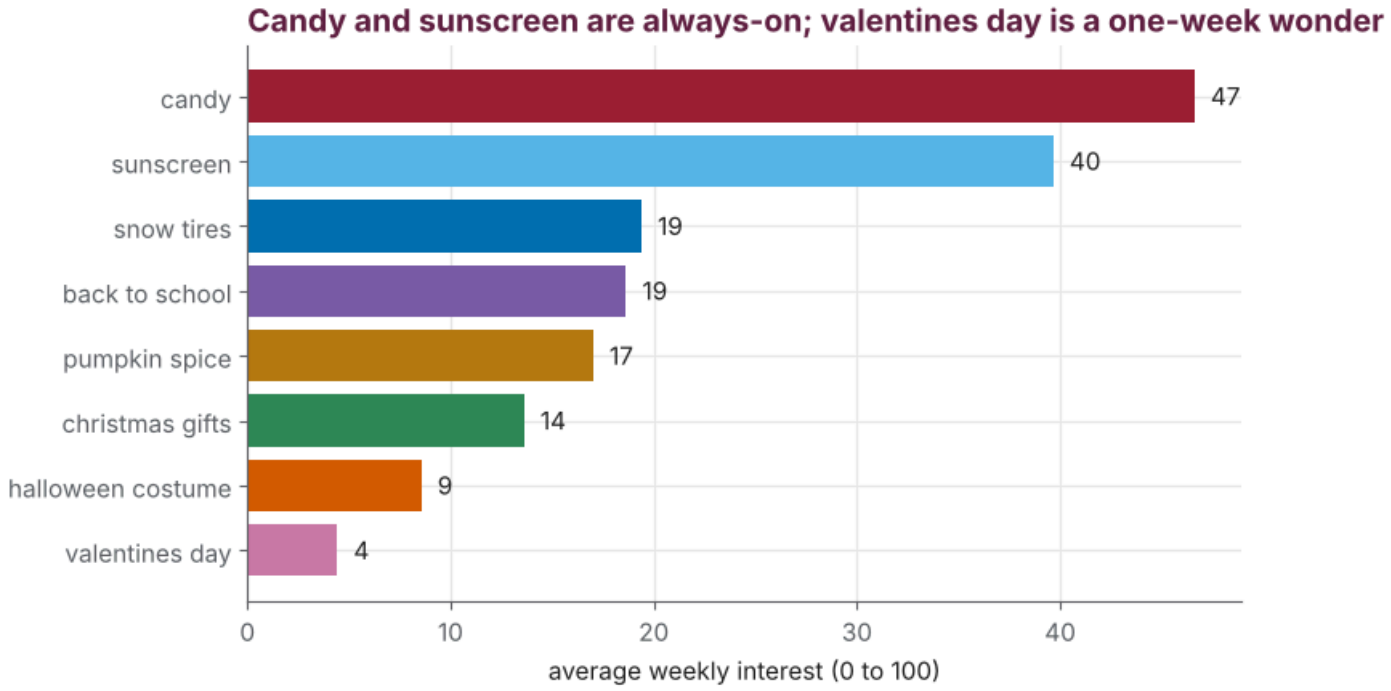


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

Costume interest is near zero for ten months, climbs through September and explodes in the last two weeks of October, the same shape every year. Candy never goes quiet; it bumps at Halloween and Christmas. Limit: relative interest, not sales.

Chart 2

Bar: which lanes are always-on and which are spikes?

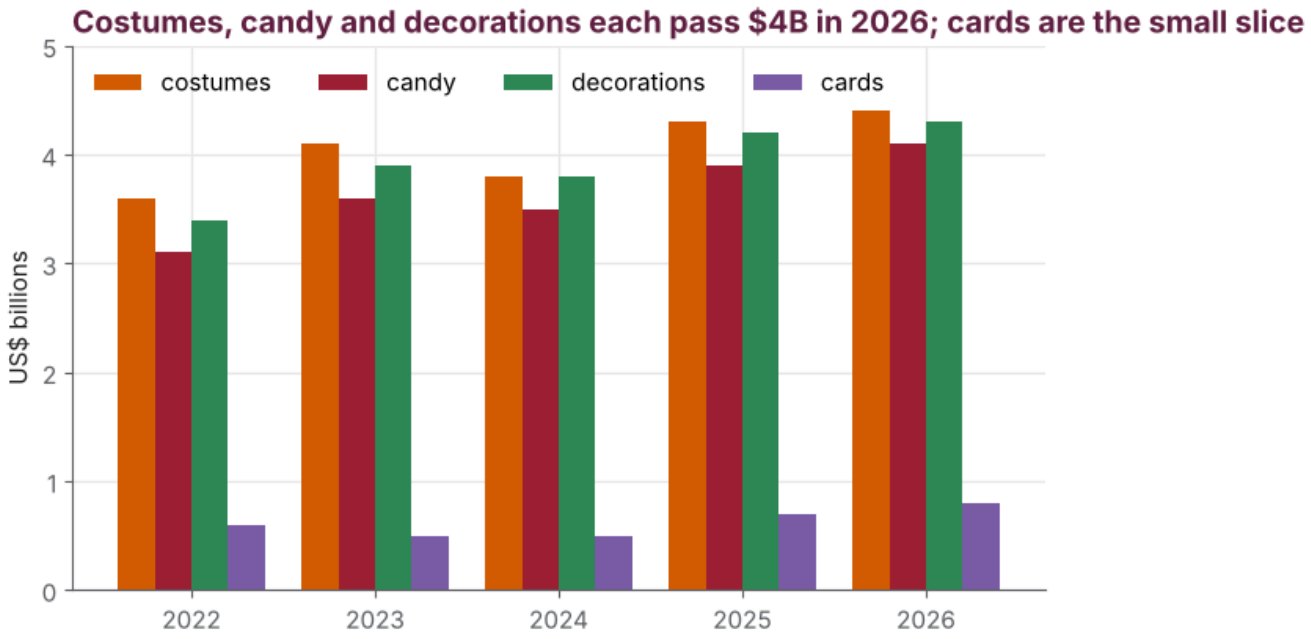


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

Candy and sunscreen sell for months; valentines day and halloween costume are almost all spike. A store in a spike lane lives or dies on timing. Limit: an average hides when.

Chart 3

Grouped bar: where does the Halloween dollar go?

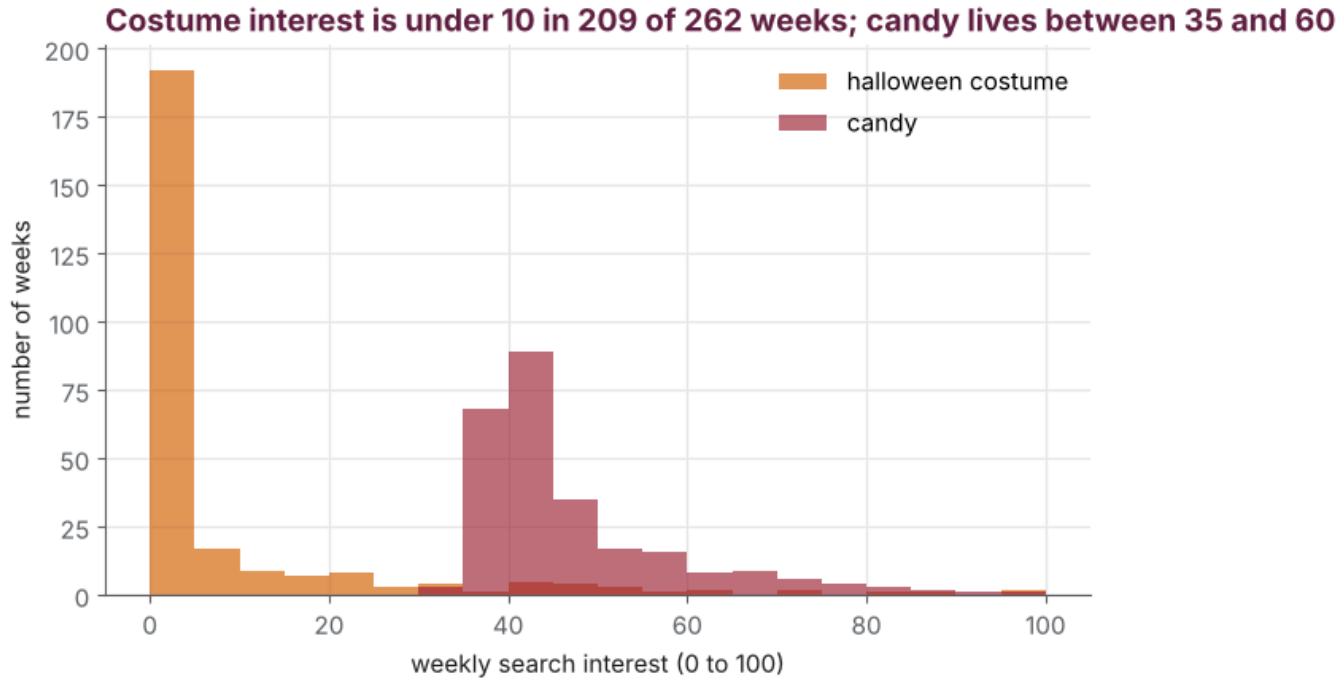


Source: NRF / Prosper Insights & Analytics annual Halloween survey (US consumers, US dollars; 2026 is the September forecast)

Costumes, candy and decorations are each about $4B in 2026, all growing, with a dip in 2024. Limit: US shoppers, US dollars, a survey forecast.

Chart 4

Histogram: how many weeks a year are actually busy?

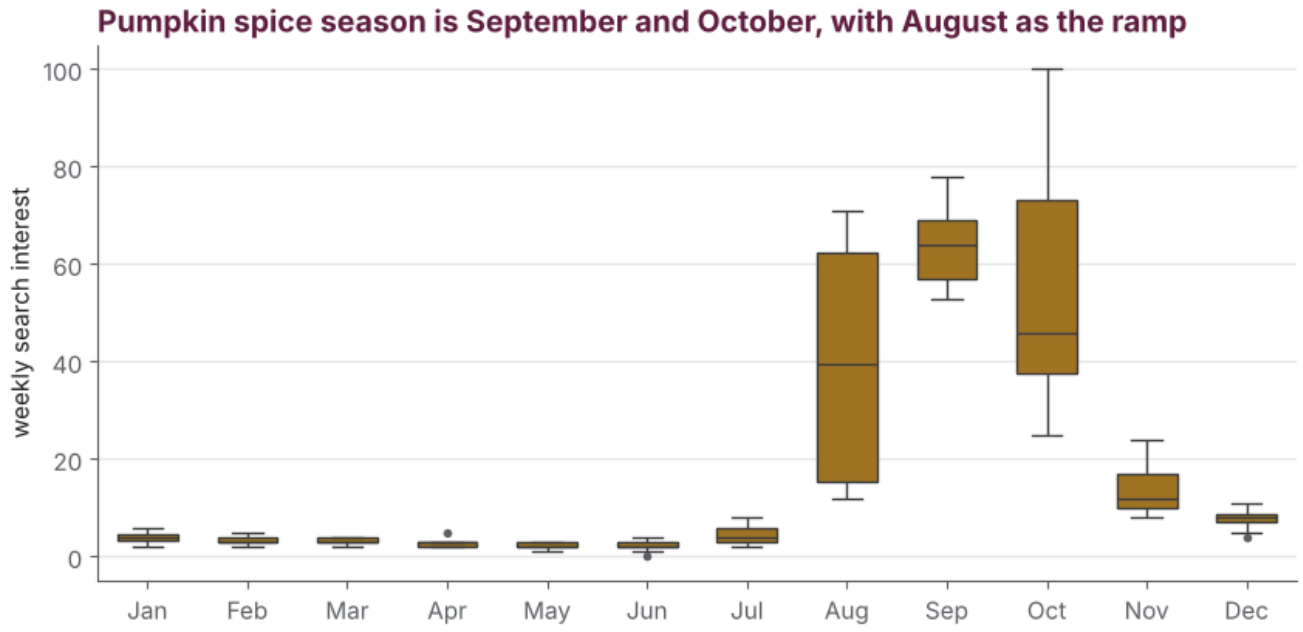


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

Costume interest is under 10 in 209 of 262 weeks and above 50 in only 12; candy lives between 35 and 60. The shape of a season. Limit: bin width is a choice (we used 5).

Chart 5

Box plot: which month, and how sure?

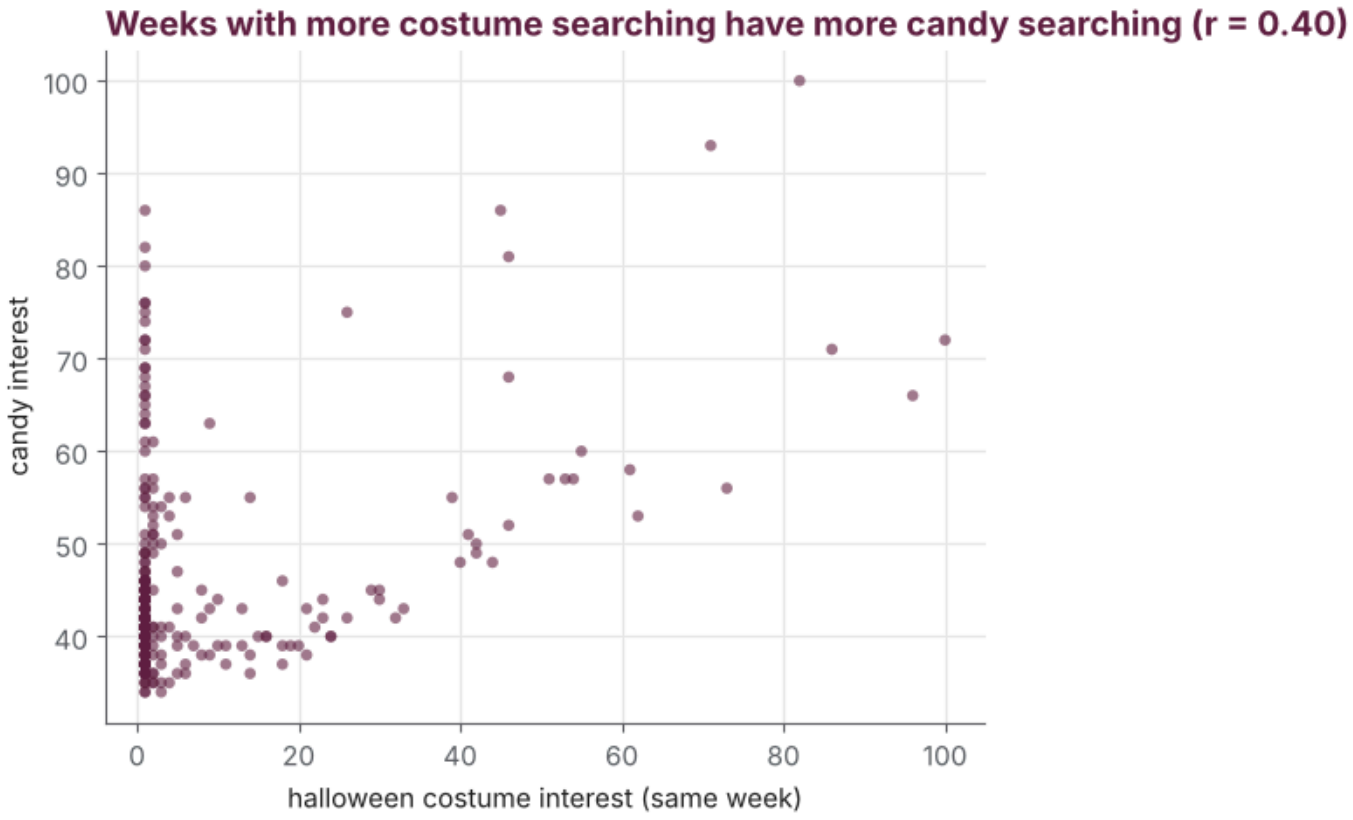


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

Pumpkin spice season is September and October; August's tall whisker is the stock-up week. Limit: five Augusts is a small sample.

Chart 6

Scatter: do costume and candy searches move together?

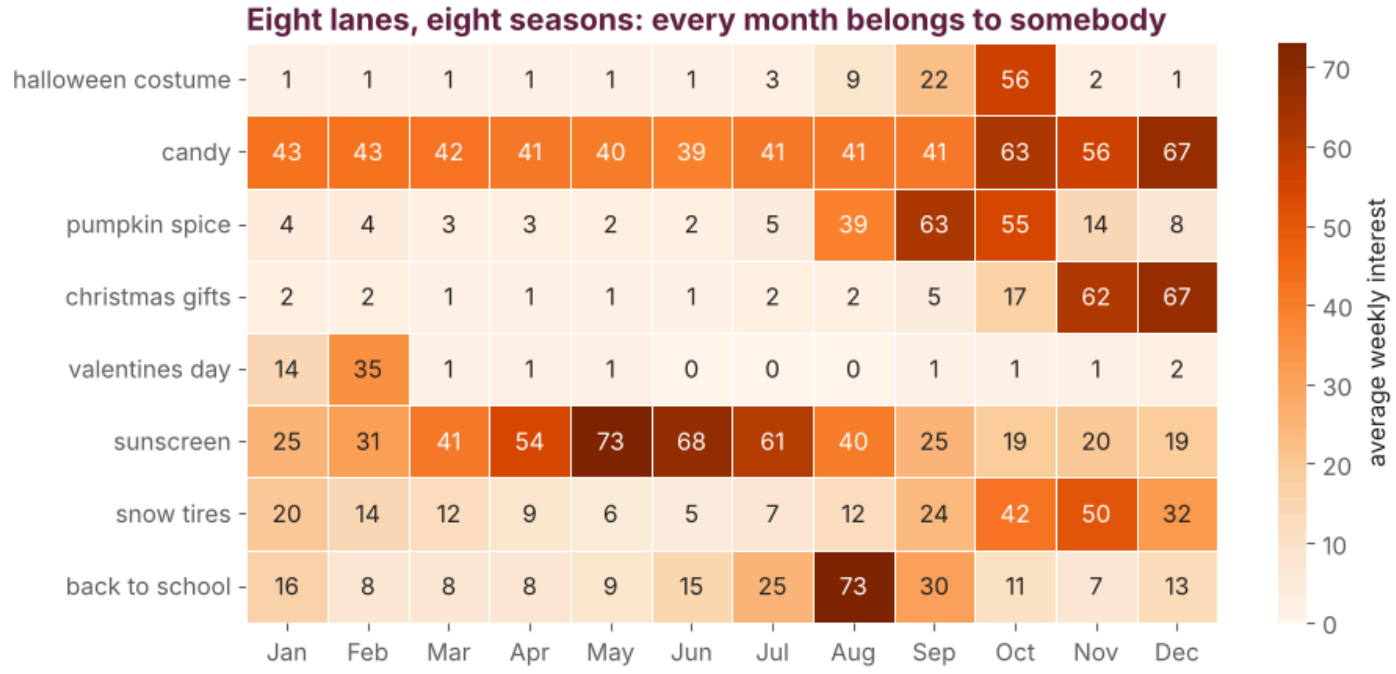


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

r = 0.40, a moderate link: busy costume weeks are only some of candy's busy weeks (candy has Christmas too). Not causation: the calendar causes both.

Chart 7

Heatmap: the whole seasonal calendar

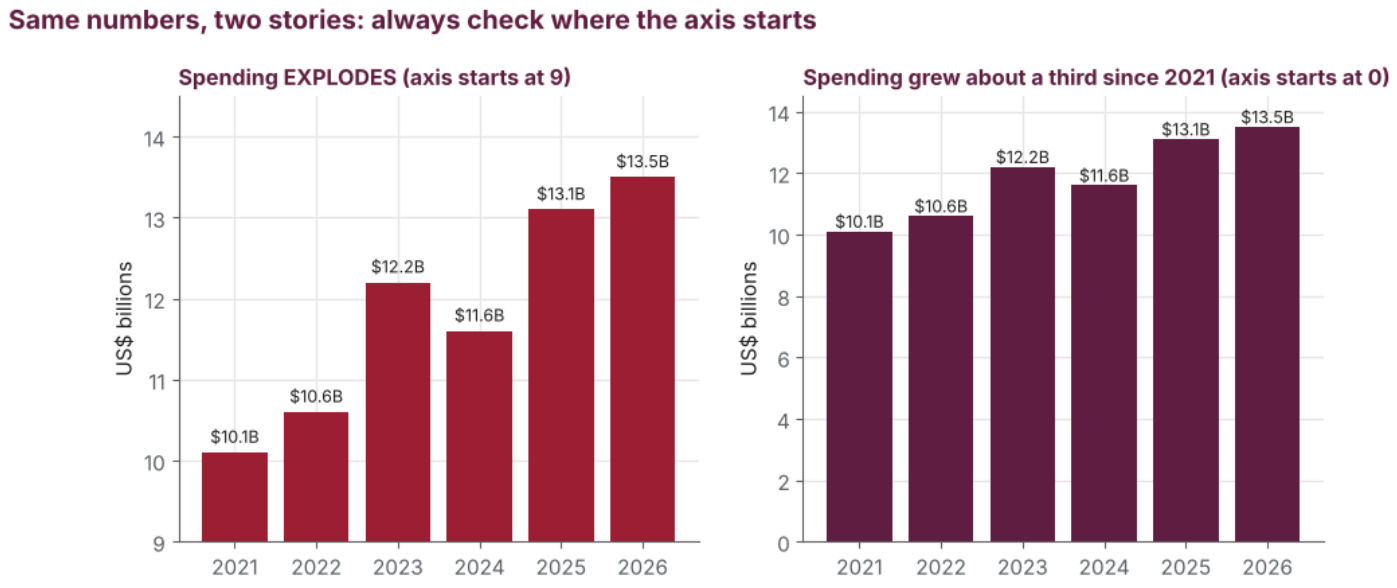


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

Read across a row for a lane's season, down a column for who is busy this month. Every month belongs to somebody. Limit: monthly averages blur the exact week.

Chart 8

Honest axis: the same numbers, two stories



Source: NRF / Prosper Insights & Analytics annual Halloween survey (US consumers, US dollars; 2026 is the September forecast)

An axis starting at 9 makes 34% growth look like an explosion; from zero it is about a third. Bars start at zero.

Season maths

Halloween costume: peak week, stock-up week, last busy week, season length (50% rule)

	lane	year	weeks_of_data	full_year	peak_week	peak	stock_up_week	last_busy_week	season_weeks
0	halloween costume	2021	10	False	2021-10-24	100	2021-10-24	2021-10-24	1
1	halloween costume	2022	52	True	2022-10-23	96	2022-10-09	2022-10-23	3
2	halloween costume	2023	53	True	2023-10-22	73	2023-10-08	2023-10-29	4
3	halloween costume	2024	52	True	2024-10-27	71	2024-10-06	2024-10-27	4
4	halloween costume	2025	52	True	2025-10-26	82	2025-10-12	2025-10-26	3
5	halloween costume	2026	43	False	2026-10-25	86	2026-10-11	2026-10-25	3

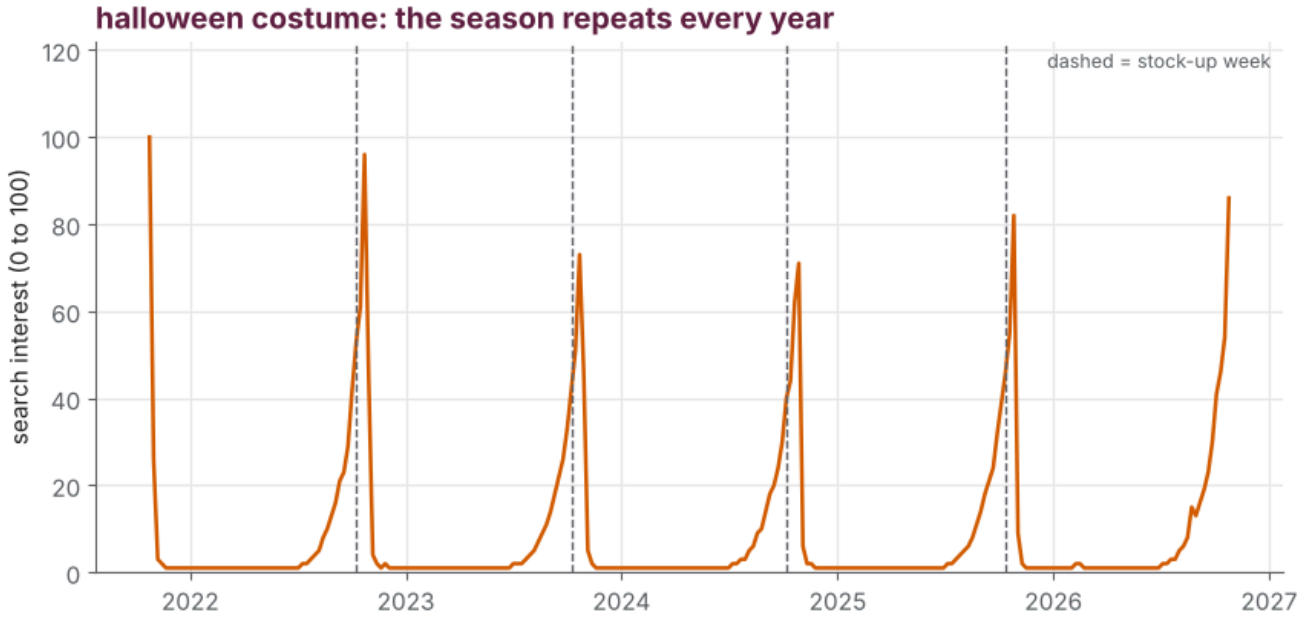
Every lane: shortest, longest and average season in weeks (full years 2022 to 2025)

	min	max	mean
lane			
halloween costume	3	4	3.5
candy	12	16	14.0
pumpkin spice	8	10	9.0
christmas gifts	6	7	6.2
valentines day	1	2	1.2
sunscreen	15	23	18.5
snow tires	2	11	7.5
back to school	5	5	5.0

Stock-up week = walking backwards from the peak, the first week still at or above 50% of it. Season = the stretch of weeks around the peak that stay above that bar. The threshold is a choice.

Section 6 · halloween costume · Step 6a

Your lane over five years, stock-up weeks marked

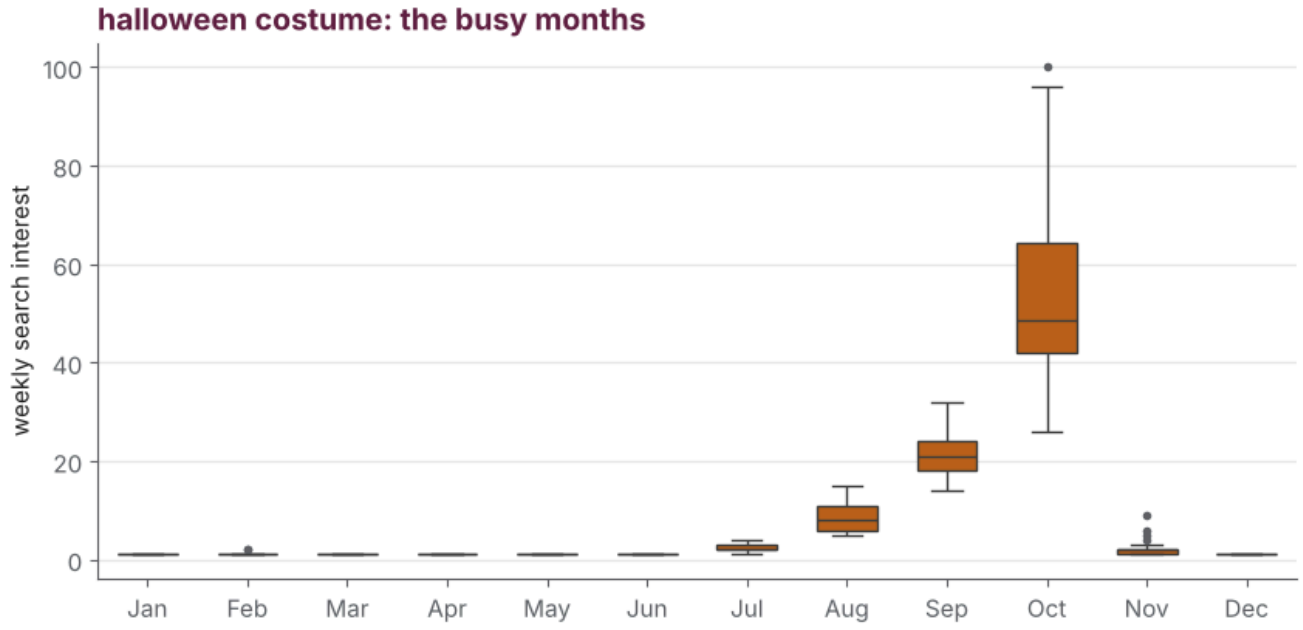


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

```
Saved lane line - halloween costume.png
```

Section 6 · halloween costume · Step 6b

Your lane by month

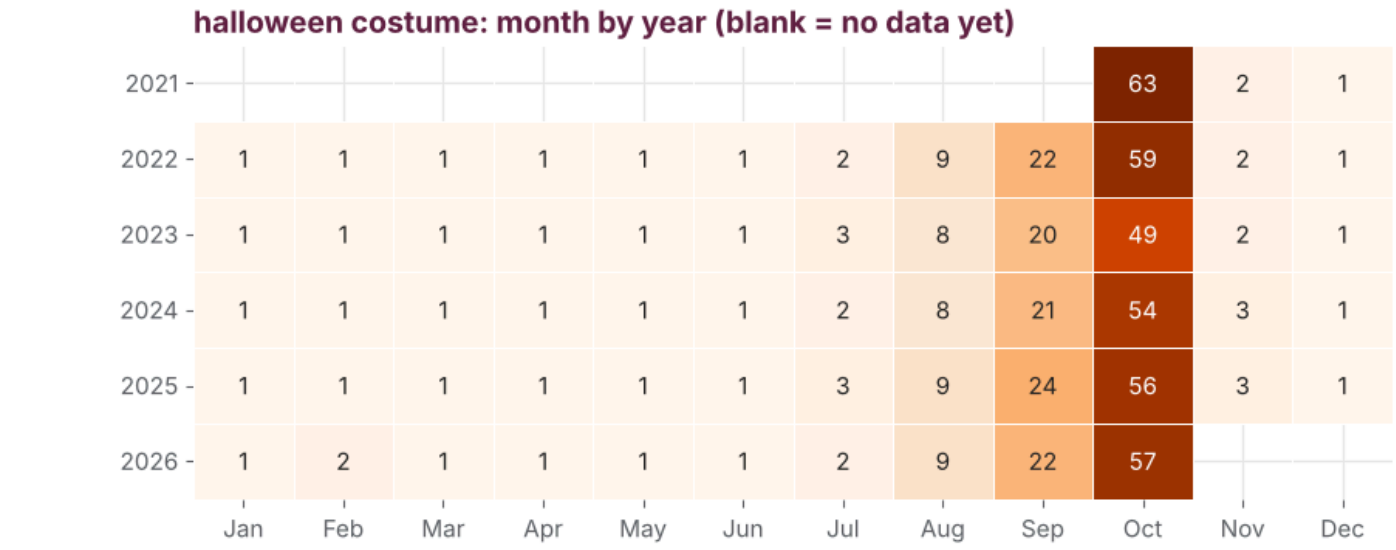


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

```
Saved lane box - halloween costume.png
```

Section 6 · halloween costume · Step 6c

Your lane month by year

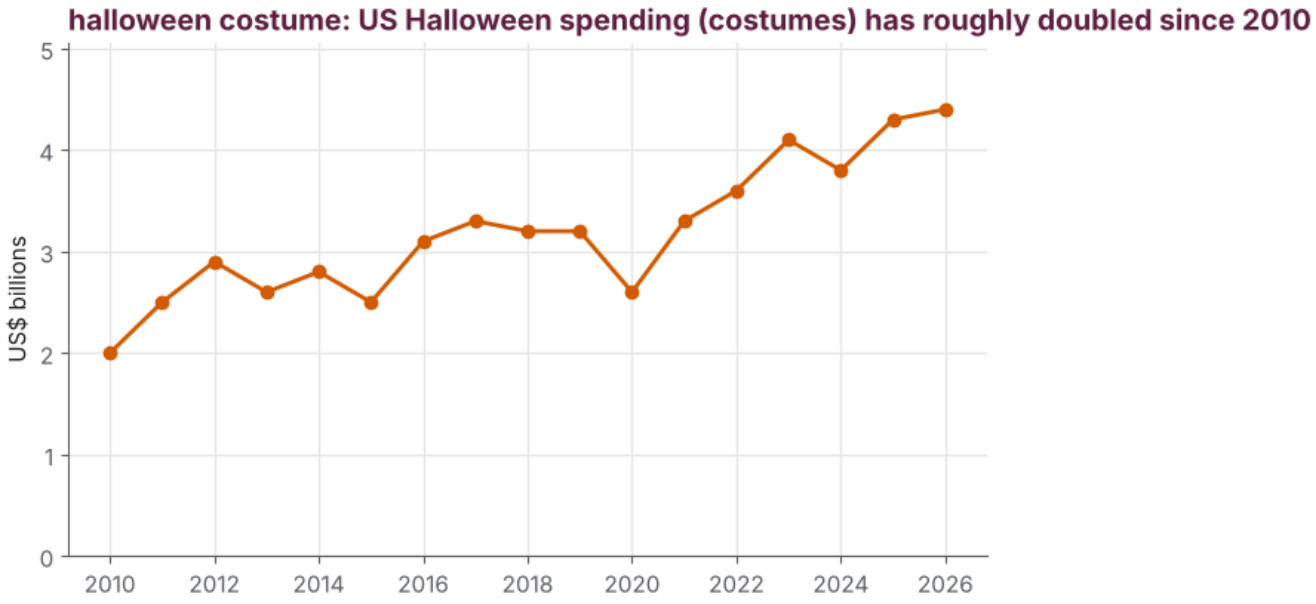


Source: Google Trends, Canada, web search, weekly, Oct 2021 to Oct 2026 (relative interest, 0 to 100 per lane)

```
Saved lane heatmap - halloween costume.png
```

Section 6 · halloween costume · Step 6e

Halloween spending for the lane



Source: NRF / Prosper Insights & Analytics annual Halloween survey (US consumers, US dollars; 2026 is the September forecast)

```
Saved lane spending - halloween costume.png
 year  costumes_billion_usd  per_person_usd  percent_celebrating
 2022                   3.6          100.45                   69
 2023                   4.1          108.24                   73
 2024                   3.8          103.63                   72
 2025                   4.3          114.45                   73
 2026                   4.4          115.14                   74
```

For an AI-built dashboard

Paste one of these small tables (not the raw data) with Dashboard Skill.md.

Season table (step 6d / 6f)

	lane	year	weeks_of_data	full_year	peak_week	peak	stock_up_week	last_busy_week	season_weeks
0	halloween costume	2021	10	False	2021-10-24	100	2021-10-24	2021-10-24	1
1	halloween costume	2022	52	True	2022-10-23	96	2022-10-09	2022-10-23	3
2	halloween costume	2023	53	True	2023-10-22	73	2023-10-08	2023-10-29	4
3	halloween costume	2024	52	True	2024-10-27	71	2024-10-06	2024-10-27	4
4	halloween costume	2025	52	True	2025-10-26	82	2025-10-12	2025-10-26	3
5	halloween costume	2026	43	False	2026-10-25	86	2026-10-11	2026-10-25	3

Month table (step 6f)

	month	mean	min	max
0	Jan	1.0	1	1
1	Feb	1.1	1	2
2	Mar	1.0	1	1
3	Apr	1.0	1	1
4	May	1.0	1	1
5	Jun	1.0	1	1
6	Jul	2.5	1	4
7	Aug	8.6	5	15
8	Sep	21.8	14	32
9	Oct	55.6	26	100
10	Nov	2.4	1	9
11	Dec	1.0	1	1

> Saved season table - halloween costume.csv and month table - halloween costume.csv · open the 📁 panel on the left to download them